Optimal Retirement Withdrawal Strategies Under Stochastic Returns and Tax Constraints

Jerry Cubbin*

March 6, 2026

Abstract

We present a framework for choosing annual withdrawals from three retirement accounts — a municipal-bond account, a traditional IRA, and a Roth IRA — to maximize sustainable after-tax spending over a fixed planning horizon. Equity returns are stochastic; the municipal-bond rate is deterministic. Because traditional IRA withdrawals are subject to federal income tax and to Required Minimum Distributions beginning at age 73, the withdrawal schedule has both tax and regulatory consequences. We formulate the single-path problem as a Linear Program (LP) and embed it in a Monte Carlo simulation over $S = 1{,}000$ independently drawn return sequences. The optimizer targets a *constant* annual after-tax consumption level C^*, eliminating year-to-year spending volatility while respecting all account-balance, tax, and RMD constraints. Under a representative calibration — \$500k muni, \$2M traditional IRA, \$1M Roth, ages 59 to 92 — the optimizer achieves a median $C^* \approx$ \$258,000 per year, roughly 7% above the median of two heuristic benchmark strategies based on fixed account-priority orderings. Median lifetime federal taxes total approximately \$560k.

*This paper benefited significantly from working with Claude Code.

Contents

1 Introduction **3**

2 Model **4**

2.1 Accounts 4

2.2 Returns and Account Dynamics 5

2.3 Income 5

2.4 Federal Income Tax 6

2.5 Required Minimum Distributions 6

2.6 Early Withdrawal Penalty 7

3 Optimization Problem **8**

3.1 Objective: Maximum Constant Consumption 8

3.2 Constraints 8

3.3 LP Reformulation 9

4 Monte Carlo Simulation **10**

5 Benchmark Strategies **11**

6 Results **12**

6.1 Baseline Calibration 12

6.2 Consumption Outcomes 12

6.3 Tax Outcomes 13

6.4 Account Drawdown Paths 15

7 Conclusion **16**

A Complete LP in Standard Form **18**

B 2026 Federal Income Tax Schedule (Married Filing Jointly) **19**

C IRS Uniform Lifetime Table (Selected Ages) **19**

1 Introduction

A retiree who holds wealth across multiple accounts faces a coordination problem that is easy to understate. The three most common account types — a taxable account invested in municipal bonds, a traditional IRA or 401(k), and a Roth IRA — differ sharply in their tax treatment: municipal-bond interest is federally tax-free, traditional IRA withdrawals are fully taxable as ordinary income and subject to Required Minimum Distributions (RMDs) beginning at age 73, and Roth withdrawals are tax-free with no withdrawal requirements. The order and timing of withdrawals therefore determines not just how much is spent each year, but how much of that spending is eroded by taxes and whether the accounts survive to the planned horizon.

Common practitioner guidance — deplete taxable accounts first, then the traditional IRA, then the Roth — ignores the interaction between withdrawal timing, marginal tax rates, RMD obligations, and the stochastic growth of account balances. In good equity markets, the Roth account can grow dramatically; in bad markets, the traditional IRA may be forced down by RMDs faster than expected. A strategy that is "optimal" on average may perform poorly on specific return paths, and the only honest assessment is a distributional one.

This paper develops a computationally tractable framework for the full three-account problem. The central design choices are as follows.

1. **Constant consumption.** We maximize C^*, the highest level of annual after-tax spending that can be sustained identically in every year of the planning horizon. The constant-consumption objective captures the preference for smooth spending and produces a single, easily interpreted number characterizing each simulated path.

2. **Linear Program per path.** For a given realization of stock returns, the piecewise-linear tax function makes the withdrawal problem a Linear Program (LP) solvable in roughly 25–40 ms with the open-source HiGHS solver. This enables $S = 1{,}000$ Monte Carlo paths in under a minute on a standard laptop.

3. **Benchmark comparison.** We evaluate the optimizer against two "account-priority" heuristics — one drawing the muni account first, the other drawing the traditional IRA first — computed with perfect foresight on each realized path

via numerical bisection. These benchmarks provide a transparent lower bound against which the value of joint optimization can be measured.

The remainder of the paper is organized as follows. Section 2 develops the model: account dynamics, returns, income, tax, and RMD constraints. Section 3 formulates the optimization problem and derives the LP reduction. Section 4 describes the Monte Carlo simulation. Section 5 defines the benchmark strategies. Section 6 presents results under baseline calibration. Section 7 concludes with a discussion of extensions. Appendix A provides the complete LP in standard matrix form; Appendix B reproduces the 2026 tax schedule; Appendix C gives the relevant portion of the IRS Uniform Lifetime Table.

2 Model

2.1 Accounts

An individual of age a_0 holds three accounts distinguished by their federal income tax treatment. Let $t = 0, 1, \ldots, T-1$ index planning years, where $T = a_\dagger - a_0$ and $a_\dagger$ is the assumed death age. Account balances A_t^j ($j \in \{m, d, r\}$) are measured at the *start* of year t, before that year's withdrawal.

- **Municipal-bond account** ($j = m$). A taxable account invested in municipal bonds. Federal law exempts muni-bond interest from income tax, so both the investment return and withdrawals from this account are federally tax-free. The account earns a fixed annual return r^m, reflecting the predictable nature of investment-grade municipal debt.

- **Traditional IRA / 401(k)** ($j = d$). A tax-*deferred* account. Contributions were made pre-tax; every dollar withdrawn is treated as ordinary income in the year of withdrawal and taxed accordingly. This account is also subject to Required Minimum Distributions starting at age $a_{\text{RMD}} = 73$ (Section 2.5).

- **Roth IRA** ($j = r$). A tax-*exempt* account. Contributions were made with after-tax dollars; qualified withdrawals — including all investment gains — are permanently tax-free. Unlike the traditional IRA, the Roth carries no RMD requirement during the account holder's lifetime, making it the most flexible account from a tax-planning standpoint.

2.2 Returns and Account Dynamics

Each account holds a blend of municipal bonds and equities. Let $\tilde{r}_t \sim \mathcal{N}(\mu, \sigma^2)$ denote the realized stock return in year t, drawn independently across years. The blended annual return on account j is

$$r_t^j \;=\; \alpha_j\, r^m \;+\; (1-\alpha_j)\,\tilde{r}_t, \qquad j \in \{d, r\}, \tag{1}$$

where $\alpha_j \in [0,1]$ is the fraction of account j held in muni bonds. The muni account itself is entirely in bonds ($\alpha_m = 1$), so its return r^m is non-stochastic.

Let $w_t^j \geq 0$ denote the withdrawal from account j in year t. Account balances evolve as

$$A_{t+1}^j \;=\; \big(A_t^j - w_t^j\big)\big(1 + r_t^j\big), \qquad t = 0, \ldots, T-1, \tag{2}$$

with given initial balances A_0^m, A_0^d, $A_0^r \geq 0$.

It is convenient to define cumulative growth factors

$$G_0^j = 1, \qquad G_{t+1}^j = G_t^j \cdot \big(1 + r_t^j\big), \tag{3}$$

so that G_t^j is the factor by which a dollar invested at time 0 in account j grows by year t (absent any withdrawals). Unrolling recursion (2) gives the closed-form balance

$$A_t^j \;=\; A_0^j\, G_t^j \;-\; \sum_{\tau=0}^{t-1} w_\tau^j \cdot \frac{G_t^j}{G_\tau^j}, \tag{4}$$

which expresses A_t^j as a linear function of the withdrawal history $\{w_0^j, \ldots, w_{t-1}^j\}$. This linearity is the key structural property exploited in the LP formulation (Section 3.3).

2.3 Income

Beyond account withdrawals, the individual receives Social Security benefits, may earn employment income before retirement, and may receive pension income:

$$y_t^{SS} = \bar{y}^{SS} \cdot \mathbf{1}\big[a_0 + t \geq a_{SS}\big], \tag{5}$$

$$y_t^{emp} = \bar{y}^{emp} \cdot \mathbf{1}\big[a_0 + t < a_{\text{ret}}\big], \tag{6}$$

$$y_t^{pen} = \bar{y}^{pen} \cdot \mathbf{1}\big[a_0 + t \geq a_{\text{ret}}\big], \tag{7}$$

where $\bar{y}^{SS}$ is the elected annual Social Security benefit, a_{SS} is the claiming age, $\bar{y}^{emp}$ is annual pre-retirement earned income, $\bar{y}^{pen}$ is the annual pension benefit, and a_{ret} is the age at which employment income ceases and pension income begins. Social Security income is partially taxable ($\phi = 0.85$); employment and pension income are fully taxable. Municipal-bond interest and Roth withdrawals are excluded from ordinary income. The taxable ordinary income in year t is therefore

$$I_t \;=\; w_t^d + \phi\, y_t^{SS} + y_t^{emp} + y_t^{pen}. \tag{8}$$

Pension income y_t^{pen} is fully included in ordinary income beginning at retirement age; setting $\bar{y}^{pen} = 0$ recovers the no-pension baseline.

2.4 Federal Income Tax

We model federal income tax under the 2026 Married Filing Jointly (MFJ) brackets, reproduced in Appendix B. The annual tax liability is computed after applying the standard deduction Δ: taxable income equals $\max(0,\, I_t - \Delta)$.

With K brackets having lower bounds L_k and marginal rates $\tau_0 < \tau_1 < \cdots < \tau_{K-1}$, the tax function can be written as

$$\mathcal{T}(I) \;=\; \sum_{k=0}^{K-1} \Delta\tau_k \cdot \max\bigl(0,\; I - \Delta - L_k\bigr), \tag{9}$$

where $\Delta\tau_k = \tau_k - \tau_{k-1}$ (with $\tau_{-1} \equiv 0$) is the incremental rate at each bracket threshold. This *incremental-rate* formulation decomposes the tax calculation into K separate "income above threshold" terms, each individually convex and non-decreasing. The function $\mathcal{T}(I)$ is piecewise linear, convex, and continuous.

Remark 1. *The incremental-rate formula (9) is algebraically equivalent to the standard "fill from the bottom" bracket calculation. Its value for our purposes is that each* $\max(0, \cdot)$ *term can be replaced by a non-negative slack variable in the LP, yielding a fully linear model (Section 3.3).*

2.5 Required Minimum Distributions

Federal law requires holders of traditional IRA and 401(k) accounts to withdraw a minimum amount each year beginning at age $a_{\text{RMD}} = 73$. The Required Minimum

Distribution (RMD) is calculated by dividing the account balance by the IRS Uniform Lifetime Table distribution period ν_t, which depends on the account holder's age (Appendix C):

$$\rho_t \;=\; \frac{A_t^d}{\nu_t}. \tag{10}$$

The RMD imposes a lower bound on traditional IRA withdrawals:

$$w_t^d \;\geq\; \rho_t \;=\; \frac{A_t^d}{\nu_t}, \qquad \text{for } a_0 + t \geq a_{\text{RMD}}. \tag{11}$$

Roth accounts are not subject to RMD requirements.

Two features of the RMD constraint are worth noting. First, because A_t^d depends on prior withdrawals, the constraint is not a simple lower bound on w_t^d in isolation; it couples the decision variables across time in a way that must be handled carefully in the LP (Section 3.3). Second, since the distribution period ν_t declines with age, the required fraction of the account that must be distributed rises over time, from roughly 3.8% at age 73 to over 8% by age 90.

2.6 Early Withdrawal Penalty

Federal law imposes a 10% penalty on traditional IRA distributions taken before the account holder reaches age $59\frac{1}{2}$. Defining the per-year penalty rate

$$\pi_t \;=\; \begin{cases} 0.10 & \text{if } a_0 + t < 59.5, \\ 0 & \text{otherwise,} \end{cases} \tag{12}$$

the total tax and penalty burden on traditional IRA withdrawals in year t is $\mathcal{T}(I_t) + \pi_t\, w_t^d$.

Because $\pi_t\, w_t^d$ is already linear in the decision variable w_t^d, the penalty integrates into the LP without introducing any new variables or constraints: it simply reduces the net consumption coefficient on w_t^d from 1 to $(1 - \pi_t)$ in the constant-consumption equality rows (Section 3.3). The optimizer therefore automatically avoids traditional IRA withdrawals in penalty-eligible years, preferring tax-free muni and Roth draws instead.

3 Optimization Problem

3.1 Objective: Maximum Constant Consumption

After-tax consumption in year t is total income from all sources minus federal income tax and any early withdrawal penalty:

$$c_t \;=\; w_t^m + w_t^d + w_t^r + y_t^{SS} + y_t^{emp} + y_t^{pen} - \mathcal{T}(I_t) - \pi_t\, w_t^d, \tag{13}$$

where π_t is defined in (12). Equivalently, $c_t = w_t^m + (1-\pi_t)\, w_t^d + w_t^r + y_t^{SS} + y_t^{emp} + y_t^{pen} - \mathcal{T}(I_t)$. Our primary objective is to find withdrawal schedules $\{w_t^m, w_t^d, w_t^r\}_{t=0}^{T-1}$ that maximize a constant annual after-tax consumption level C^* subject to $c_t = C^*$ for every year t. In other words, we look for the highest sustainable flat spending rate over the entire planning horizon.

The constant-consumption objective has several practical advantages. It directly quantifies the "permanent income" the portfolio can support, yielding a single scalar C^* that is easy to interpret and compare across simulations. It also eliminates the need to specify a discount rate or utility function: maximizing the level of a permanently flat consumption stream requires no preference assumptions beyond the preference for more over less.

3.2 Constraints

For a given realized return path $\{\tilde{r}_t\}_{t=0}^{T-1}$, the optimization problem is subject to the following constraints.

(i) **Non-negative withdrawals.** $w_t^j \geq 0$ for all $j \in \{m, d, r\}$ and $t = 0, \ldots, T-1$. Withdrawals cannot be negative; re-contribution of funds is not modeled.

(ii) **Non-negative account balances.** $A_t^j \geq 0$ for all j and $t = 1, \ldots, T$. An account cannot be overdrawn. Using representation (4), this condition becomes a set of linear inequalities in the withdrawal history up to time t.

(iii) **RMD constraint.** Substituting the closed-form expression (4) for A_t^d into constraint (11) gives

$$\nu_t\, w_t^d + \sum_{\tau=0}^{t-1} w_\tau^d \cdot \frac{G_t^d}{G_\tau^d} \;\geq\; A_0^d\, G_t^d, \qquad a_0 + t \geq a_{\mathrm{RMD}}, \tag{14}$$

which is linear in the withdrawal variables.

(iv) **Constant consumption.** $c_t = C^*$ for all $t = 0, \ldots, T-1$, where C^* is treated as a scalar decision variable to be maximized. After the tax linearization described in Section 3.3, this yields T linear equality constraints.

Remark 2 (Terminal balances and feasibility)**.** *We do* not *impose terminal equality constraints $A_T^j = 0$. The balance non-negativity constraints already enforce $A_T^j \geq 0$, and the maximize-C^* objective naturally pushes all three accounts toward zero by incentivizing higher withdrawals throughout the horizon. In practice, the muni and traditional accounts are depleted to zero in the large majority of simulations.*

The Roth account, held entirely in equities, can grow faster than the constant-consumption constraint allows it to be drawn down — particularly in Social Security years when SS income already covers most of C^*, leaving little room for additional Roth withdrawals without exceeding C^*. On such paths, a small positive Roth balance remains at death: an unintended bequest rather than an infeasibility.

Imposing a terminal equality $A_T^j = 0$ instead would render these extreme paths infeasible with no optimal solution. The inequality formulation is always feasible.

3.3 LP Reformulation

The tax function $\mathcal{T}(I_t)$ in constraint (iv) is piecewise linear and convex, so it cannot appear directly in an LP. We handle it by introducing *bracket slack variables* $s_{tk} \geq 0$ defined to satisfy

$$s_{tk} \;=\; \max\bigl(0,\; I_t - \Delta - L_k\bigr) \;=\; \max\bigl(0,\; w_t^d + \phi\, y_t^{SS} + y_t^{emp} + y_t^{pen} - \Delta - L_k\bigr). \quad (15)$$

With this definition, the tax on year-t income is exactly

$$\mathcal{T}(I_t) \;=\; \sum_{k=0}^{K-1} \Delta\tau_k \cdot s_{tk}, \quad (16)$$

which is linear in the slack variables. The constraint $s_{tk} \geq I_t - \Delta - L_k$ is imposed as the LP inequality

$$w_t^d - s_{tk} \;\leq\; \Delta + L_k - \phi\, y_t^{SS} - y_t^{emp} - y_t^{pen}, \qquad t = 0, \ldots, T-1,\; k = 0, \ldots, K-1. \quad (17)$$

Because the LP objective drives s_{tk} to its minimum feasible value (higher slacks represent higher taxes, which reduce the consumption that can be achieved), the slacks will equal exactly the quantities in (15) at any optimal solution.

Substituting (16) into the constant-consumption condition $c_t = C^*$ and rearranging using (13) yields

$$w_t^m + (1-\pi_t)\, w_t^d + w_t^r - \sum_{k=0}^{K-1} \Delta\tau_k\, s_{tk} - C^* \;=\; -y_t^{SS} - y_t^{emp} - y_t^{pen}, \qquad t = 0, \ldots, T-1, \tag{18}$$

which is a system of T linear equalities in the decision variables. The coefficient $(1-\pi_t)$ on w_t^d reflects the early withdrawal penalty: in years before age $59\frac{1}{2}$ only 90% of each traditional IRA dollar withdrawn reaches after-tax consumption.

Proposition 1. *The optimal withdrawal problem is a Linear Program in the $N+1 = 3T + TK + 1$ variables*

$$\big(\, w_0^m, \ldots, w_{T-1}^m,\;\; w_0^d, \ldots, w_{T-1}^d,\;\; w_0^r, \ldots, w_{T-1}^r,\;\; \{s_{tk}\},\;\; C^* \,\big),$$

with objective $\min\; -C^*$*, subject to the linear constraints* (17) *and* (18)*, the balance non-negativity constraints derived from* (4)*, the RMD constraints* (14)*, and the non-negativity of all withdrawal and slack variables. The variable C^* is unrestricted in sign.*

Appendix A presents the full LP in standard matrix form. Under baseline calibration ($T = 33$, $K = 7$) the LP has $3(33) + 33(7) + 1 = 331$ variables and approximately 600 constraints, and is solved in 25–40 ms by HiGHS.

4 Monte Carlo Simulation

For each simulation $s = 1, \ldots, S$, we independently draw a T-year stock-return sequence $\{\tilde{r}_t^{(s)}\}_{t=0}^{T-1}$, with each $\tilde{r}_t^{(s)} \sim \mathcal{N}(\mu, \sigma^2)$. All paths use the same mean μ and volatility σ; the randomness enters only through the realized draws. A fixed random seed is used to ensure reproducibility.

For each path we:

1. Compute the blended returns r_t^d and r_t^r and their cumulative growth factors G_t^d and G_t^r (the muni growth factors G_t^m are path-independent).

2. Construct the LP described in Section 3.3.
3. Solve the LP to obtain the optimal withdrawals $\{w_t^{m*}, w_t^{d*}, w_t^{r*}\}$ and constant consumption $C^{*(s)}$.
4. Compute account balance paths A_t^j, taxes $\mathcal{T}(I_t)$, and effective marginal rates from the solution.
5. Compute both benchmark consumption levels $C^{\mathrm{mrd}(s)}$ and $C^{\mathrm{drm}(s)}$ by bisection (Section 5).

All results are stored in a long-format panel indexed by (s, t) and summarized across simulations by quantiles of $C^{*(s)}$, lifetime taxes $\sum_t \mathcal{T}(I_t^{(s)})$, and account balance trajectories.

5 Benchmark Strategies

To assess the value of joint optimization, we compare C^* against two heuristic strategies that observe the same realized return path but follow a fixed account-priority ordering rather than solving an LP.

Muni-first (C^{mrd}): $A^m \to A^r \to A^d$

Draw first from the municipal-bond account until it is exhausted, then from the Roth, then from the traditional IRA. In each year, the withdrawal from the currently active account is the minimum needed to reach the target consumption C, netting any taxes owed on the traditional IRA portion. The ordering was chosen to defer taxable income as long as possible, letting the tax-deferred and Roth accounts compound on a larger base.

Given a realized path, the maximum sustainable constant consumption under this ordering is found by bisection: increase C until the terminal balance of the last account (traditional IRA) transitions from positive to negative, then recover the zero-crossing by Brent's method.

Deferred-first (C^{drm}): $A^d \to A^r \to A^m$

Draw first from the traditional IRA until exhausted, then the Roth, then the muni account. This front-loads the tax burden but leaves the tax-free accounts to grow

for longer. The same bisection procedure is applied, with the terminal muni-account balance as the target function.

Properties of the benchmarks

Both strategies are always feasible (account balances are floored at zero rather than required to hit exactly zero), and both use perfect foresight over the realized return path. The optimizer has strictly more freedom: it can mix withdrawals across accounts in any year, is not constrained to a priority ordering, and jointly optimizes the tax consequences of all decisions. Therefore $C^{*(s)} \geq \max(C^{\mathrm{mrd}(s)}, C^{\mathrm{drm}(s)})$ for every path s, and the gap between C^* and the benchmarks measures the value of joint optimization relative to the best simple heuristic.

6 Results

6.1 Baseline Calibration

Table 1 summarizes the parameters used in the baseline simulation. The planning horizon spans $T = 33$ years (ages 59–91). The individual has no employment income ($\bar{y}^{emp} = 0$), retires at age 65 (though this has no effect in the baseline since $\bar{y}^{emp} = 0$), claims Social Security at age 70, and faces RMDs from age 73 onward. The Roth is held entirely in equities ($\alpha_r = 0$) to maximize tax-free compounding; the traditional IRA holds a 40% muni bond position ($\alpha_d = 0.40$) to moderate volatility on the taxable account. All 1,000 simulated paths are solved to optimality.

6.2 Consumption Outcomes

Table 2 reports the cross-simulation distribution of annual after-tax consumption under the optimizer and the two benchmarks.

The optimizer achieves a median constant consumption of \$258,249 per year, approximately 6.6% above the muni-first benchmark (\$242,254) and 8.2% above the deferred-first benchmark (\$238,590). The improvement arises from the optimizer's ability to blend withdrawals across accounts in every year to keep the traditional IRA's marginal contribution to taxable income within lower brackets. In particular, the optimizer draws more heavily from the traditional IRA in pre-Social Security

Table 1: Baseline Parameters

Parameter	**Symbol**	**Value**
Ages		
Start age	a_0	59
Retirement age	a_{ret}	65
Social Security age	a_{SS}	70
RMD start age	a_{RMD}	73
Death age	$a_\dagger$	92
Initial account balances		
Municipal-bond account	A_0^m	\$500,000
Traditional IRA	A_0^d	\$2,000,000
Roth IRA	A_0^r	\$1,000,000
Asset allocation (muni fraction)		
Traditional IRA	α_d	40%
Roth IRA	α_r	0%
Investment returns		
Municipal-bond rate	r^m	3.0%
Stock mean return	μ	7.0%
Stock volatility	σ	15.0%
Social Security		
Annual benefit	$\bar{y}^{SS}$	\$120,000
Taxable fraction	ϕ	85%
Simulation		
Number of paths	S	1,000
Tax schedule		2026 MFJ (Appendix B)

years (ages 59–69), when $I_t = w_t^d$ alone and the standard deduction absorbs the first \$32,200 of income, then draws more from the tax-free muni and Roth accounts in later years when Social Security adds \$102,000 of taxable income. The fixed-ordering heuristics cannot make this intertemporal adjustment.

Figure 1 shows the distributional comparison between the optimizer and the two benchmarks.

6.3 Tax Outcomes

Table 3 reports the cross-simulation distribution of lifetime federal income taxes.

Table 2: Distribution of Annual After-Tax Consumption Across 1,000 Simulations ($)

Strategy	**P10**	**P25**	**Median**	**P75**	**P90**
Optimizer (C^*)	211,803	230,654	258,249	294,341	336,231
Muni-first (C^{mrd})	197,900	216,148	242,254	270,799	301,866
Deferred-first (C^{drm})	195,588	213,816	238,590	268,612	302,104

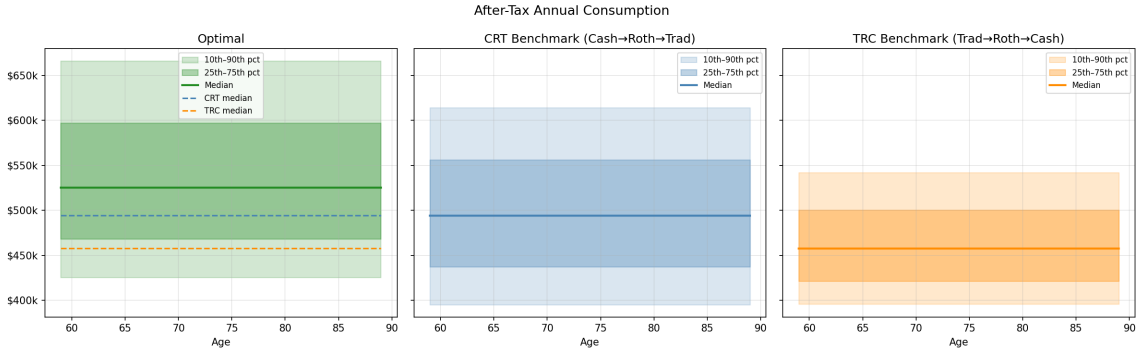

Figure 1: After-tax annual consumption: optimizer (left), muni-first benchmark C^{mrd} (center), deferred-first benchmark C^{drm} (right). Bands show 10th–90th and 25th–75th percentiles across 1,000 simulations; the solid line is the median. Dotted lines in the left panel overlay the benchmark medians for reference.

Median lifetime taxes are $560,429 over 33 years, or roughly $17,000 per year on median constant consumption of $258,249 — an effective lifetime rate of approximately 6.6%. The P90/P10 spread of $239,089 is driven by variation in stock returns: stronger equity performance inflates the traditional IRA balance and forces larger taxable withdrawals, generating higher taxes even under an optimal withdrawal policy. This cross-simulation variance is irreducible, reflecting the fundamental link between portfolio outcomes and tax obligations in a tax-deferred account.

Figure 2 shows the distribution of effective marginal tax rates by age. The optimizer consistently keeps most paths in the 12–22% brackets throughout the horizon, avoiding the 24%+ brackets by moderating traditional IRA withdrawals in years when Social Security is also included in taxable income.

Table 3: Distribution of Lifetime Federal Income Taxes Across 1,000 Simulations ($)

	P10	**P25**	**Median**	**Mean**	**P75**	**P90**
Lifetime taxes	462,802	504,819	560,429	575,430	630,523	701,891

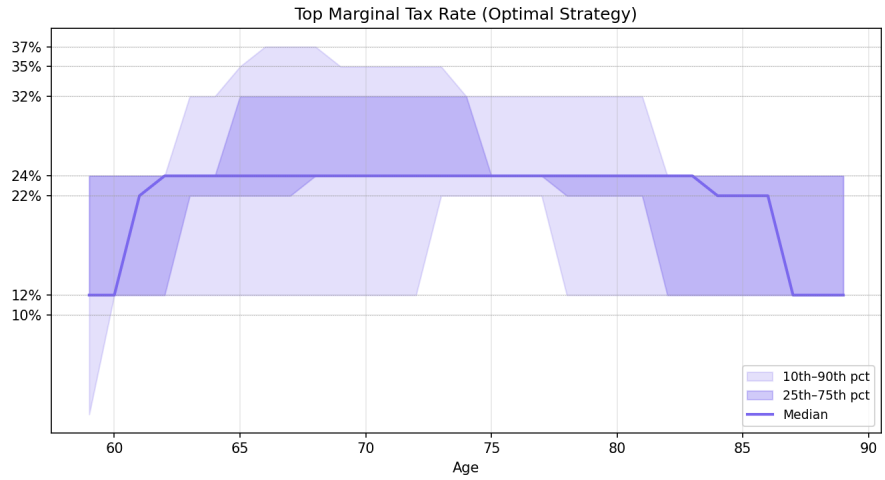

Figure 2: Distribution of marginal federal tax rates by age across 1,000 simulations.

6.4 Account Drawdown Paths

Figure 4 shows the median and interquartile range of each account balance over the planning horizon; Figure 5 shows the corresponding annual withdrawals.

Several patterns are visible. First, the traditional IRA is drawn down substantially in the pre-Social Security years (ages 59–69), when the sole source of taxable income is w_t^d itself and the marginal rate is relatively low. As Social Security begins at age 70 and adds $102,000 to taxable income, traditional IRA withdrawals fall and the remaining traditional IRA balance is primarily liquidated through RMDs. Second, the muni account serves as a bridge, supplementing traditional IRA income before Social Security begins. Third, the Roth account is drawn down later, once the other tax-free muni source is exhausted.

On high-return paths, the Roth account retains a small positive balance at death (median terminal Roth $\approx$ $140,000). As discussed in Remark 2, this reflects the constraint that Social Security income in later years already covers most of C^*, leaving limited capacity to draw additional tax-free income from the Roth without simply accepting that it will not be fully depleted.

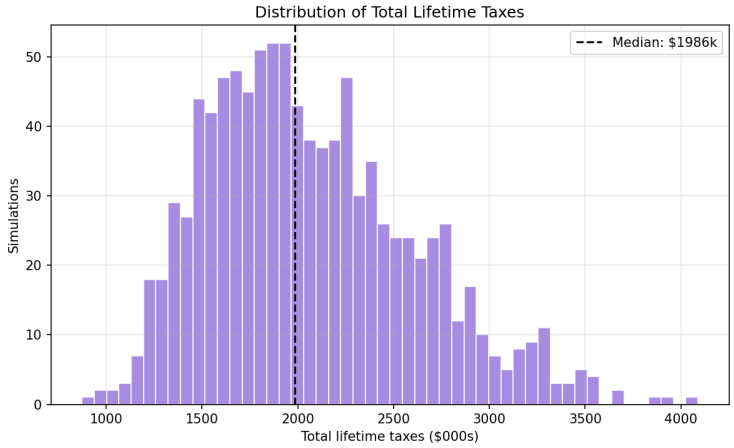

Figure 3: Cross-simulation distribution of lifetime federal income taxes.

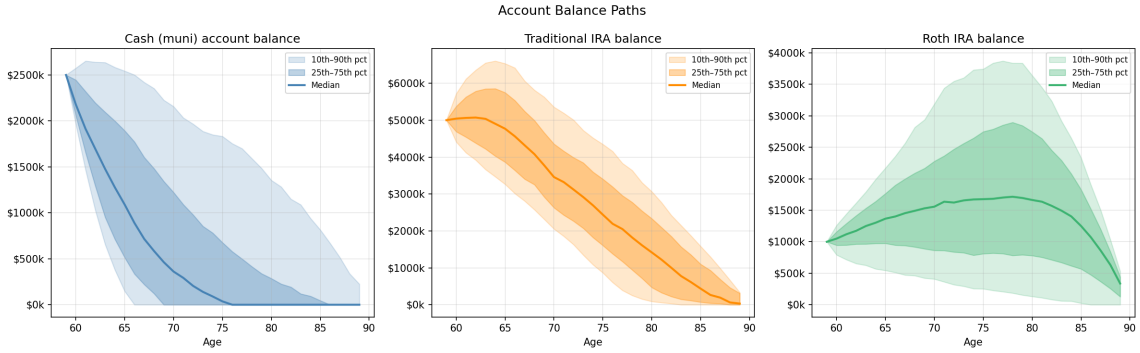

Figure 4: Account balances by age: muni-bond account (left), traditional IRA (center), Roth IRA (right). Bands show 10th–90th and 25th–75th percentiles; solid line is the median.

7 Conclusion

We have developed and implemented a Monte Carlo–LP framework for optimizing retirement account withdrawals across three account types with distinct tax treatments. The core observation is that, for a given realized sequence of stock returns, the constant-consumption withdrawal problem is a Linear Program with roughly 300 variables and 600 constraints, solvable in milliseconds. A Monte Carlo wrapper over 1,000 independent return paths yields a complete distributional picture of sustainable consumption and lifetime taxes.

Under our baseline calibration — $3.5 million in initial wealth, $120,000 in annual Social Security benefits, and historical equity return parameters — the optimizer

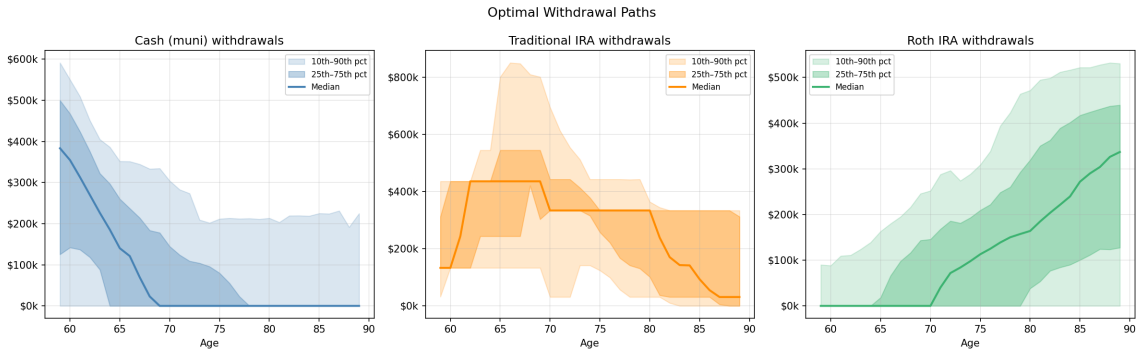

Figure 5: Annual withdrawals by account type. Muni and Roth withdrawals are tax-free; traditional IRA withdrawals constitute ordinary income.

supports a median constant after-tax consumption of \$258,249 per year, roughly 7% above simple account-priority heuristics. Lifetime federal taxes are held to a median of \$560,000 over 33 years, with effective marginal rates concentrated in the 12–22% range throughout the horizon.

Several extensions are natural and left for future work. First, the Social Security claiming age a_{SS} is treated as fixed; an outer optimization over a_{SS} could find the claiming age that maximizes expected C^*. Second, the model could incorporate state income taxes, which vary significantly across states and affect the relative attractiveness of muni versus traditional IRA withdrawals. Third, the planning horizon $a_\dagger$ could be treated as stochastic, integrating over a mortality distribution rather than conditioning on a fixed death age. Fourth, Roth conversion strategies — transferring funds from the traditional IRA to the Roth prior to RMD age to reduce future taxable distributions — could be incorporated as an additional set of decision variables.

A Complete LP in Standard Form

We index the decision variables as follows. Let T be the horizon length and $K = 7$ the number of tax brackets. Define index functions

$$\begin{aligned}
i_m(t) &= t, && t = 0, \ldots, T-1, \\
i_d(t) &= T + t, && t = 0, \ldots, T-1, \\
i_r(t) &= 2T + t, && t = 0, \ldots, T-1, \\
i_s(t,k) &= 3T + tK + k, && t = 0, \ldots, T-1,\ k = 0, \ldots, K-1, \\
i_C &= 3T + TK. && (\text{scalar } C^*)
\end{aligned}$$

The total number of variables is $N + 1 = 3T + TK + 1$. Let $\mathbf{x} \in \mathbb{R}^{N+1}$ collect all variables.

Objective vector

$\mathbf{c} \in \mathbb{R}^{N+1}$: all entries zero except $c_{i_C} = -1$.

Equality constraints

One row per year $t = 0, \ldots, T-1$, enforcing constant after-tax consumption:

$$x_{i_m(t)} + (1 - \pi_t)\, x_{i_d(t)} + x_{i_r(t)} - \sum_{k=0}^{K-1} \Delta\tau_k\, x_{i_s(t,k)} - x_{i_C} \;=\; -y_t^{SS} - y_t^{emp} - y_t^{pen}.$$

The coefficient $(1-\pi_t)$ on $x_{i_d(t)}$ incorporates the early withdrawal penalty (12); $\pi_t = 0$ for all years in which the planner is age $59\frac{1}{2}$ or older.

Inequality constraints

All inequalities are of the form $\mathbf{A}_{\text{ub}}\,\mathbf{x} \le \mathbf{b}_{\text{ub}}$.

(a) Tax bracket slacks (TK rows):

$$x_{i_d(t)} - x_{i_s(t,k)} \;\le\; \Delta + L_k - \phi\, y_t^{SS} - y_t^{emp} - y_t^{pen}, \quad t = 0, \ldots, T-1,\ k = 0, \ldots, K-1.$$

(b) Muni-account non-negativity (T rows, for $t = 1, \ldots, T$):

$$\sum_{\tau=0}^{t-1} \frac{G_t^m}{G_\tau^m} x_{i_m(\tau)} \;\leq\; A_0^m \, G_t^m.$$

(c) Traditional IRA non-negativity (T rows):

$$\sum_{\tau=0}^{t-1} \frac{G_t^d}{G_\tau^d} x_{i_d(\tau)} \;\leq\; A_0^d \, G_t^d.$$

(d) Roth non-negativity (T rows):

$$\sum_{\tau=0}^{t-1} \frac{G_t^r}{G_\tau^r} x_{i_r(\tau)} \;\leq\; A_0^r \, G_t^r.$$

(e) RMD constraints (one row per t with $a_0 + t \geq a_{\text{RMD}}$):

$$-\nu_t \, x_{i_d(t)} - \sum_{\tau=0}^{t-1} \frac{G_t^d}{G_\tau^d} x_{i_d(\tau)} \;\leq\; -A_0^d \, G_t^d.$$

Variable bounds

$$x_{i_m(t)},\; x_{i_d(t)},\; x_{i_r(t)},\; x_{i_s(t,k)} \;\geq\; 0; \qquad x_{i_C} \in \mathbb{R} \text{ (unrestricted).}$$

B 2026 Federal Income Tax Schedule (Married Filing Jointly)

Taxable income is ordinary income I_t less the standard deduction: $\max(0,\, I_t - \$32{,}200)$. The tax function implemented in the optimizer is $\mathcal{T}(I) = \sum_k \Delta\tau_k \cdot \max(0, I - \Delta - L_k)$, where the bracket increments are $\Delta\tau_0 = 10\%$, $\Delta\tau_1 = 2\%$, $\Delta\tau_2 = 10\%$, $\Delta\tau_3 = 2\%$, $\Delta\tau_4 = 8\%$, $\Delta\tau_5 = 3\%$, $\Delta\tau_6 = 2\%$.

C IRS Uniform Lifetime Table (Selected Ages)

The distribution period ν_t in the RMD calculation is taken from the IRS Uniform Lifetime Table (Rev. Proc. 2022-32). This table applies to unmarried owners and to

Table 4: 2026 Married Filing Jointly Tax Brackets

Taxable income: from	**to**	**Marginal rate**
\$0	\$24,800	10%
\$24,800	\$100,800	12%
\$100,800	\$211,400	22%
\$211,400	\$403,550	24%
\$403,550	\$512,450	32%
\$512,450	\$768,700	35%
\$768,700	∞	37%
Standard deduction (MFJ 2026)		\$32,200

married owners whose spouse is not more than 10 years younger. The full table used in the optimizer covers ages 60 through 120.

Table 5: IRS Uniform Lifetime Table: Selected Ages

Age	**Distribution period ν**
73	26.5
74	25.5
75	24.6
76	23.7
77	22.9
78	22.0
79	21.1
80	20.2
85	16.0
90	12.2
95	8.9

The required minimum distribution for a given year is $\rho_t = A_t^d/\nu_t$. At age 73 this requires distributing approximately 3.8% of the account balance; by age 90 the required fraction exceeds 8%.